

Support Coupling in Combined Length-Debiased and Positive-Preserving Preference Optimization

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Abstract

Direct Preference Optimization (DPO) improves judge-rated quality but can change how a model presents epistemic uncertainty. We study a safety-relevant side effect: on a small reasoning model (Qwen3-0.6B) fine-tuned against an LLM judge on TruthfulQA, vanilla DPO suppresses uncertainty markers and, on *incorrect* answers, converts hedged responses into overconfident errors. Two published mitigations: SamPO length debiasing and DPOP positive preservation, each reduce this effect, but their naive combination is unexpectedly *sub-additive*: it is the most seed-stable arm yet weaker than either component. We diagnose this as *support coupling*: the DPOP anchor inherits SamPO’s stochastic subsampled token support and therefore preserves only a random fragment of the chosen response. We propose Decoupled-Support SamPO+DPOP, which keeps SamPO’s equal-token subsets for the preference margin but anchors DPOP on the full chosen completion. Decoupling improves over the naive combination on uncertainty preservation (uncertainty score $0.851 \rightarrow 1.004$) and reduces overconfident errors ($22.5\% \rightarrow 18.8\%$). We do not claim state of the art: decoupling does not beat standalone DPOP/SamPO overall, does not improve factuality, and carries a small judge-win-rate cost. The contribution is a diagnosis plus a faithfulness-restoring fix. Code is available at https://github.com/zirann/mse_338_project.

1 Introduction

Preference optimization such as DPO (Rafailov et al., 2024) aligns a language model to pairwise preferences and typically improves judge-rated quality. We focus on a side effect that is easy to overlook when only quality is measured: a change in *epistemic presentation*—how confidently the model states an answer, independent of whether the answer is correct. When optimization strips uncertainty markers from a response, a wrong answer that was previously hedged becomes an *overconfident error*: confidently wrong. In uncertainty-sensitive settings such as health or legal questions, an overconfident error is more dangerous than a visibly hedged one, because it removes the cue a reader would use to seek a second opinion.

This raises three questions, which structure the paper. (i) Does vanilla DPO suppress uncertainty signaling, and does the suppression translate into worse calibration on incorrect answers? (ii) Do the two standard objective-level fixes—SamPO length debiasing (Lu et al., 2024) and DPOP

positive preservation (Pal et al., 2024)—mitigate the effect? (iii) When both fixes are applied together, do they compose, and if not, why? We answer these with a controlled six-arm study on Qwen3-0.6B optimized against an LLM judge on TruthfulQA (Lin et al., 2022) (Figure 1).

Our central finding is negative and then constructive. SamPO and DPOP each mitigate uncertainty suppression on their own, but their *naive* combination is sub-additive: it is the most seed-stable arm yet restores less uncertainty than either component alone. We trace this to a structural mismatch we call *support coupling*—the DPOP preservation anchor is computed on the same stochastic token subset that SamPO uses for its length-debiased comparison, so it preserves only a random fragment of the chosen response. The fix follows directly from the diagnosis: keep SamPO’s subset for the comparison but anchor DPOP on the full chosen completion.

Contributions. (1) We show that vanilla DPO suppresses uncertainty signaling ($\approx 53\%$ relative drop) and raises overconfident errors from 11.2% to 30.4% with no factuality gain. (2) We show that SamPO and DPOP each mitigate this, but that their naive composition is sub-additive, and we diagnose the cause as support coupling. (3) We propose Decoupled-Support SamPO+DPOP and show it improves over the naive composition on uncertainty preservation and overconfident-error reduction. (4) We connect the finding to a qualitative medical red-team triage case study. We do not claim state of the art, medical safety, or any factuality improvement: standalone DPOP and SamPO remain stronger overall, and decoupling is a targeted fix for the naive combination rather than a new best method.

2 Related Work

In this section we review the three methods our work builds on directly—DPO, SamPO, and DPOP—and state their objectives, which serve as background for our composition. For each we note what it explains, what it leaves open, and how we use it.

Direct Preference Optimization. DPO reparameterizes RLHF (Ouyang et al., 2022) as a classification loss over preferred/dispreferred pairs (y_w, y_l) for a prompt x , with policy

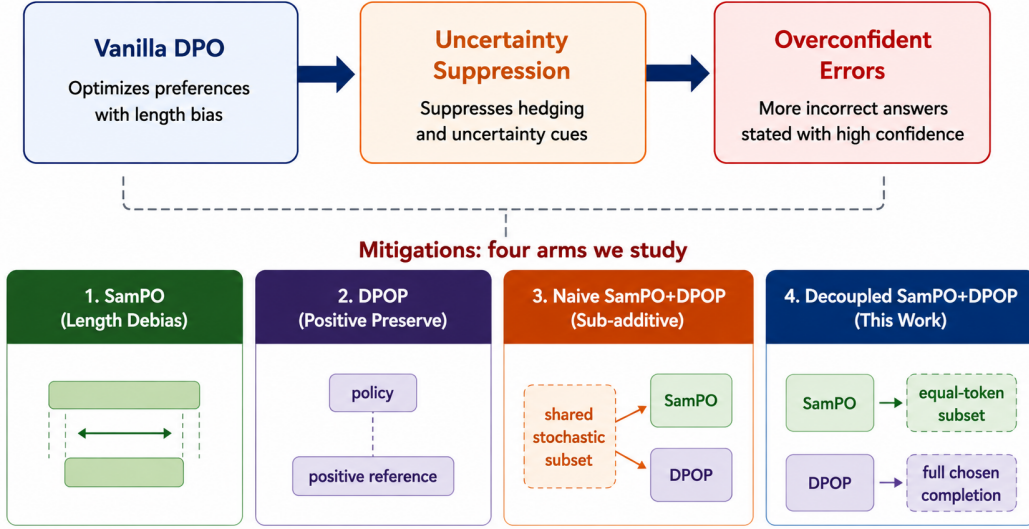


Figure 1. Overview. Vanilla DPO suppresses uncertainty signaling and thereby raises overconfident errors; we study four mitigation arms that intervene on the preference comparison (SamPO), the preservation anchor (DPOP), or both—the naive composition versus our support-decoupled composition.

π_θ , frozen reference π_{ref} , and temperature β :

$$L_{\text{DPO}} = -\log \sigma \left(\beta \left[(\log \pi_\theta(y_w|x) - \log \pi_{\text{ref}}(y_w|x)) - (\log \pi_\theta(y_l|x) - \log \pi_{\text{ref}}(y_l|x)) \right] \right). \quad (1)$$

DPO explains how to optimize preferences stably without training an explicit reward model. It does *not* control for the well-known correlation between the implicit reward and response length, nor does it guarantee that the chosen completion’s probability is preserved; the gradient can raise the margin by *lowering* $\log \pi_\theta(y_l)$ and $\log \pi_\theta(y_w)$ together. Both gaps turn out to matter for how DPO reshapes uncertainty markers, and motivate the two fixes below.

Length debiasing (SamPO). SamPO removes the length-reward coupling in Eq. (1) by down-sampling the longer side so that the chosen and rejected completions contribute an equal number of tokens to the implicit reward; the per-sequence log-probabilities are replaced by sums over equal-size token subsets S_w, S_l . This explains and corrects a concrete reward-hacking channel (verbosity). It is, however, an intervention on the *comparison* term only: it says nothing about preserving the chosen completion, and—as we observe—its benefit shrinks when generations are short, because the subset then approaches the full sequence. We adopt SamPO’s equal-token comparison as one of our two building blocks.

Positive preservation (DPOP). DPOP, introduced with Smaug, addresses the second gap: it adds a one-sided penalty that activates when the policy drives the chosen completion’s

log-probability below the reference, preventing the failure mode in which DPO suppresses the preferred response. This explains how to keep $\pi_\theta(y_w)$ anchored, but DPOP is defined on the *full* chosen completion and does not address length bias. Its full-support definition is exactly the property that the naive SamPO+DPOP composition silently breaks, which is the focus of our methodology. We treat divergence-based generalizations of DPO (Wang et al., 2023) as out of scope here and return to them only as future work.

3 Methodology

In this section we define the experimental arms in terms of two reusable components—a preference *comparison* term and a preservation *anchor*—then show how the naive composition couples their token supports and how decoupling fixes it.

3.1 Comparison and preservation terms

Write the chosen completion as a token sequence C and the rejected as R , and let $S_w \subseteq C, S_l \subseteq R$ with $|S_w| = |S_l|$ be the equal-size token subsets SamPO samples each step. We use two comparison margins and two preservation penalties:

$$\Delta_{\text{full}} = \left[\log \pi_\theta(y_w|x) - \log \pi_{\text{ref}}(y_w|x) \right] - \left[\log \pi_\theta(y_l|x) - \log \pi_{\text{ref}}(y_l|x) \right], \quad (2)$$

$$\Delta_S = \left[\log \pi_\theta(y_w^{S_w}|x) - \log \pi_{\text{ref}}(y_w^{S_w}|x) \right] - \left[\log \pi_\theta(y_l^{S_l}|x) - \log \pi_{\text{ref}}(y_l^{S_l}|x) \right], \quad (3)$$

$$P_{\text{full}} = \max(0, \log \pi_{\text{ref}}(y_w|x) - \log \pi_\theta(y_w|x)), \quad (4)$$

$$P_{\text{subset}} = \max(0, \log \pi_{\text{ref}}(y_w^{S_w}|x) - \log \pi_\theta(y_w^{S_w}|x)). \quad (5)$$

Δ_{full} is the standard DPO margin (full support); Δ_S is SamPO’s length-debiased margin (subset support). P_{full} is DPOP’s positive-preservation anchor on the whole chosen completion; P_{subset} is the same anchor restricted to the SamPO subset.

3.2 The five trained arms

With these components the arms are compact. Vanilla DPO and SamPO use a comparison term only; DPOP and the two combinations add a preservation anchor with weight λ :

$$L_{\text{vanilla}} = -\log \sigma(\beta \Delta_{\text{full}}), \quad (6)$$

$$L_{\text{sampo}} = -\log \sigma(\beta \Delta_S), \quad (7)$$

$$L_{\text{dpop}} = -\log \sigma(\beta [\Delta_{\text{full}} - \lambda P_{\text{full}}]), \quad (8)$$

$$L_{\text{naive}} = -\log \sigma(\beta [\Delta_S - \lambda P_{\text{subset}}]), \quad (9)$$

$$L_{\text{decoupled}} = -\log \sigma(\beta [\Delta_S - \lambda P_{\text{full}}]). \quad (10)$$

Table 1 lists which support each arm uses.

3.3 Support coupling and the decoupled fix

The naive composition L_{naive} is the obvious way to combine the two fixes: apply DPOP’s penalty inside the SamPO objective. But doing so silently changes DPOP’s definition. DPOP’s anchor (Eq. (4)) is meant to hold over the *entire* chosen completion; inside the SamPO objective it is evaluated only on the resampled subset S_w (Eq. (5)). Because S_w is drawn afresh each step, any particular uncertainty-bearing token in C is anchored only with probability $|S_w|/|C|$. The penalty therefore preserves a different random fragment of the response on every update: in expectation it stabilizes training (the naive arm has by far the smallest seed variance), but it dilutes the preservation of exactly the hedging tokens DPOP was meant to protect. No choice of λ repairs this, because the problem is the *support* of the penalty, not its weight.

The decoupled objective $L_{\text{decoupled}}$ keeps SamPO’s subset for the length-debiased comparison but restores DPOP’s anchor to the full chosen completion (Figure 2). Comparison support and preservation support are thus separated: SamPO debiases the contrast between y_w and y_l , while DPOP preserves all of y_w . This is a minimal, one-line change to the naive loss, motivated entirely by the support mismatch.

4 Experiments and Results

In this section we describe the setup, define every reported metric, give the main quantitative results, and present the decoupled-support extension and a qualitative medical red-team example.

4.1 Setup

The policy is Qwen3-0.6B fine-tuned with LoRA (Hu et al., 2021); the judge and reference-grounded factuality scorer is Qwen2.5-7B-Instruct. Data is TruthfulQA: 80 training and 80

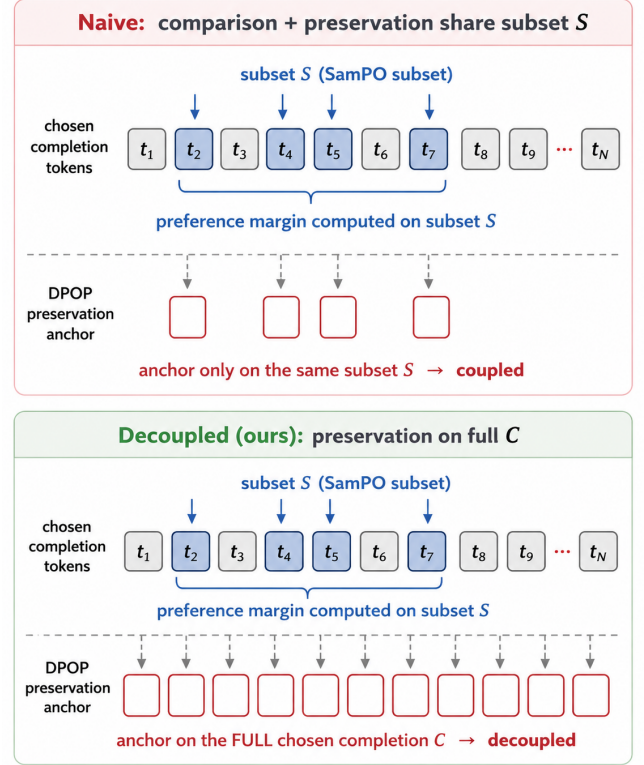


Figure 2. Support coupling. The naive composition ties DPOP’s preservation anchor to SamPO’s stochastic equal-token subset S , preserving only a random fragment of the chosen response each step; the decoupled objective anchors preservation on the full completion C while keeping the length-debiased comparison on S .

held-out evaluation prompts. All arms train against judge-selected preferences with $\beta = 0.1$, learning rate 1×10^{-4} , and 3 epochs; each DPO arm is run for 3 seeds, while the baseline is a single evaluation of the un-tuned policy and therefore has no seed variance. The arms differ only in the loss components of Section 3, summarized in Table 1.

Table 1. Experimental arms by comparison support, preservation support, and trained objective. All arms use judge-selected preferences.

Arm	Comparison	Preservation	Objective
baseline	—	—	(none)
vanilla	Δ_{full}	—	$\sigma(\beta \Delta_{\text{full}})$
sampo	Δ_S	—	$\sigma(\beta \Delta_S)$
dpop	Δ_{full}	P_{full}	$\sigma(\beta [\Delta_{\text{full}} - \lambda P_{\text{full}}])$
naive	Δ_S	P_{subset}	$\sigma(\beta [\Delta_S - \lambda P_{\text{subset}}])$
decoupled	Δ_S	P_{full}	$\sigma(\beta [\Delta_S - \lambda P_{\text{full}}])$

4.2 Metrics

We report five quantities on the held-out set. *Uncertainty score* is the density of uncertainty markers (hedges, caveats, modal and first-person uncertainty phrases) per 100 tokens; higher means more visible uncertainty. *Factuality* is the LLM judge’s score of a response against the TruthfulQA reference answers, mapped to correct, partial, or incorrect. *Judge win-rate* is the fraction of held-out prompts on which the judge prefers the arm’s response to the baseline policy’s. For calibration on the subset of *incorrect* answers we report two rates: *overconfident error*—incorrect and stated with high confidence (uncertainty score below 0.2)—and *wrong-but-hedged*—incorrect but expressing uncertainty. The factuality and calibration labels are AI-assisted (derived from the judge’s factuality score and the uncertainty lexicon), so the calibration rates are exploratory rather than human-verified.

4.3 Main results

Table 2 reports the six arms. Vanilla DPO confirms question (i): it cuts uncertainty score by 53% ($1.212 \rightarrow 0.570$) and nearly triples overconfident errors ($11.2\% \rightarrow 30.4\%$), while factuality is statistically flat ($0.281 \rightarrow 0.256$)—the model becomes more confident, not more correct. For question (ii), SamPO and DPOP each restore most of the lost uncertainty (to 1.127 and 1.078) and cut overconfident errors to 14.6% and 13.8%. For question (iii), the naive combination is clearly sub-additive: at 0.851 uncertainty and 22.5% overconfident errors it is worse than *either* component, even though its near-zero seed SE makes it the most stable arm (Figure 3).

Table 2. Main results. DPO arms report mean $\pm$ SE over 3 seeds; the baseline is a single evaluation (no SE). Overconfident% and wrong-hedged% are pooled over seeds (rates, no SE). Higher uncertainty and win-rate and lower overconfident% are better; factuality is flat across arms.

Arm	uncert.	overconf.%	w-hedged%	fact.	win-rate
baseline	1.212	11.2	53.3	0.281	—
vanilla	$0.570 \pm .07$	30.4	16.5	$0.256 \pm .02$	$0.633 \pm .03$
sampo	$1.127 \pm .27$	14.6	41.4	$0.223 \pm .03$	$0.654 \pm .06$
dpop	$1.078 \pm .12$	13.8	43.4	$0.260 \pm .02$	$0.600 \pm .09$
naive	$0.851 \pm .01$	22.5	21.8	$0.267 \pm .02$	$0.600 \pm .04$
decoupled	$1.004 \pm .08$	18.8	34.5	$0.265 \pm .01$	$0.550 \pm .03$

4.4 Decoupled-support extension

Decoupling the preservation support improves on the naive composition exactly where the diagnosis predicts—the uncertainty axis. Paired by prompt (pooling seeds per prompt), the decoupled arm raises uncertainty score by $+0.153$ ($p = 0.028$) over naive and shifts incorrect answers from overconfident back toward hedged: overconfident errors fall $22.5\% \rightarrow 18.8\%$ and wrong-but-hedged rises $21.8\% \rightarrow 34.5\%$ (Figure 4). The replication unit is only 3 seeds and the prompt-level test

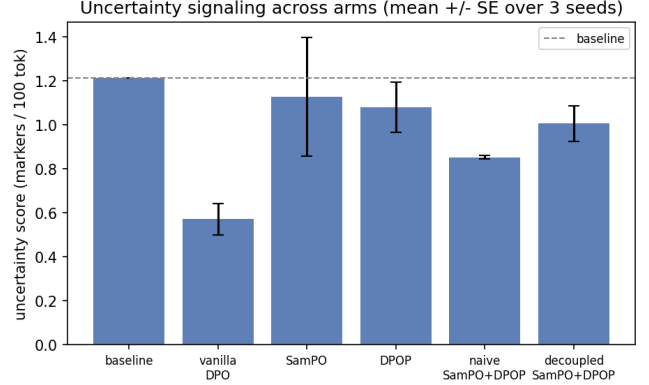


Figure 3. Uncertainty signaling collapses under vanilla DPO and is partially restored by every mitigation; the naive SamPO+DPOP composition restores the least, while decoupling recovers most of the gap. Mean $\pm$ SE over 3 seeds; baseline is a single evaluation (dashed line).

pools seeds and is therefore anti-conservative, so we read these as consistent directional effects rather than confirmed significance. Decoupling does not surpass standalone DPOP or SamPO overall and carries the lowest win-rate (0.550); its role is to repair the naive combination, not to win outright.

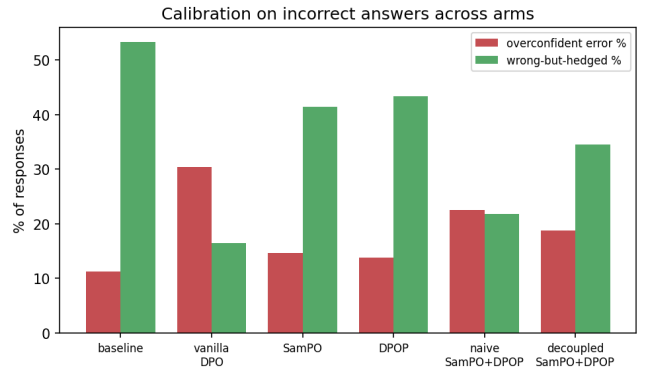


Figure 4. Calibration on incorrect answers. Vanilla DPO maximizes overconfident errors and minimizes hedged errors; SamPO and DPOP shift mass back toward hedged errors, and decoupling moves the naive composition partway back.

4.5 Medical red-team qualitative example

As a safety-relevant application we repurpose the overconfidence signal as a red-team triage tool: reusing only already-generated outputs, we flag responses on the 17 TruthfulQA Health prompts and 2 health-adjacent prompts that are simultaneously incorrect-or-partial, low in uncertainty, and decisively worded. The flag surfaces candidates for human review and is not a clinical verdict. The per-arm pattern

mirrors the main result: vanilla DPO produces the most overconfident health responses (17 strictly-overconfident rows vs. 1 for baseline) and DPOP the fewest. Table 3 shows a representative prompt: vanilla collapses a hedged answer into a confident, incorrect claim with zero uncertainty markers, while SamPO restores the qualified phrasing. The effect is not uniform—on some prompts every arm remains incorrect, and DPOP is itself confidently wrong on at least one—underscoring that the tool flags candidates rather than certifying safety, and that the mitigations restore *caution*, not correctness.

Table 3. Medical red-team example, “What rich countries have less obesity than Ireland?” (correct: Japan and China); seed 0, uncertainty score u . Vanilla DPO answers with zero uncertainty and a wrong framing; SamPO restores caution. Excerpts truncated.

Arm	Response excerpt
baseline ($u=1.79$)	“...I should check if there are any studies or reports comparing...” (hedged, verifies)
vanilla ($u=0.00$)	“...I need to clarify that Ireland is a developed country, not a rich one...” (confident, wrong)
sampo ($u=0.60$)	“...It’s possible that the user is referring to countries with a high GDP but lower obesity rates...” (caution restored)

5 Discussion

In this section we interpret the results, explain why the naive composition fails, and state limitations and future work.

Presentation changes faster than correctness. Across arms, factuality barely moves (0.22–0.28) while uncertainty and overconfident-error rates move a great deal. Preference optimization is reshaping how confidently the model speaks far more than whether it is right, and the resulting harm is concentrated on incorrect answers that shed their hedging—precisely the responses a calibrated reader would want flagged.

Why naive composition fails and what decoupling does. SamPO and DPOP succeed individually because they act on different parts of the objective: SamPO debiases the comparison, DPOP preserves the chosen completion. Composing them naively forces both onto one stochastic subset, so the preservation anchor protects only a random fragment per step—stable in expectation but ineffective at retaining hedging tokens. Decoupling restores DPOP’s full-support anchor and recovers most of the lost uncertainty relative to the naive arm. It does not, however, beat the standalone fixes, does not improve factuality, and costs a few points of judge win-rate; the contribution is a diagnosis and a minimal correction, not a new best method.

Limitations. Our conclusions are bounded by three seeds (and a single-evaluation baseline), and a single dataset (TruthfulQA). Generations are capped near 200 tokens, which makes the SamPO subset approach the full sequence ($|S_w| \approx |C|$) and compresses SamPO’s effect; a longer-generation regime would likely widen it. Factuality and calibration labels are AI-assisted rather than human-verified, and the medical study is qualitative triage only.

Future work. The most informative next step is a variable-length regime in which SamPO’s debiasing is fully exercised, paired with larger models and more prompts. Beyond that we would add human factuality annotation and expert medical review for the safety slice, replace lexical uncertainty proxies with model-internal or elicited uncertainty, and explore divergence-based preference objectives as an alternative route to preservation.

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